Data Management with `hallmark`

TIMESTEP Software Engineering Interns
2026-04-29, EHT Analysis WG Meeting

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TIMESTEP was started by Dr. Gurtina Besla, with an NSF Early Career Grant.

TIMESTEP focuses on increasing student retention in STEM fields by:

1. **Building community** and providing key career guidance workshops throughout the year
2. Improving **student access to paid research** (Apprenticeship Program)
3. Increasing access to **industry experiences** (Summer Tech Internship)
4. Engaging students in **industry best practices** in the Astro SE Internship

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TIMESTEP Software Engineering Internship

- Piloted this academic year with 4 students (2 juniors and 2 seniors)
- Weekly workshops to develop coding skills not taught in classes
- Engage in industry best practices (Stand-ups, Pragmatic Programmer, Github, Testing, etc)
- 10 hours paid each week
- Working together on a team to work on / develop Hallmark

Hoping to expand this internship to more University of Arizona students, included in US EHT Proposal (could work with your team to develop software)

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EHT Data Products

The EHT has very diverse data products, e.g., raw data from observation, correlated visibility, calibrated (fringe fitted) visibility, GRMHD, GRRT

- EHTC_2017L1_May2022
- EHTC_FirstSgrAResults_May2022
- EHTC_metadata2018_Dec2023
- EHTC_FirstSgrAPol_Mar2024
- GRMHD v3
- GRRT v3

EHT Data Products

Each dataset is kind of a database by itself, e.g.,

```
(ckc) ckchan@eht:~> tree ~/2017SGRA/model/Illinois_thermal
/bd3/home/ckchan/2017SGRA/model/Illinois_thermal
├── 230GHz
│   ├── Ma+0.5_w3
│   │   ├── img_s03000_Rh10_i10.h5
│   │   ├── img_s03000_Rh10_i110.h5
│   │   ├── img_s03000_Rh10_i130.h5
...
```

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EHT Data Products

Each dataset is kind of a database by itself, e.g.,

```
(ckc) ckchan@eht:~> tree ~/2017SGRA/model/KORAL_long
/bd3/home/ckchan/2017SGRA/model/KORAL_long
├── A_am.9_1e-5/
│   └── Rlow1.0_Rhigh20.0
│       ├── inc10.0
│       │   ├── freq140000.000000
│       │   │   ├── dump1000_time10000.0
│       │   │   │   └── A_a0_1e-5_Rlow1.0_Rhigh20.0_inc10.0_...
```

Output of a step in the pipeline may not be the ideal input for the next step

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Features Needed

The EHT (and many data heavy projects) require data management tools that can

- manage large data sets (many files and large files)
- version control
- automate data pipelining
- compatible with a large number of storage backends
- manage data sets as database

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Existing Solutions

	Manage-ment	Version control	Pipelining	Multiple backends	Meta-database
Git LFS	✓	✓	✓	✗	✗
Git Annex	✓	✓	✓	✗	✗
DVC	✓	✓	✓	✓	✗
`hallmark`	✓	✓	✓	✓	✓

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`hallmark`: Intro

https://github.com/l6a/hallmark

Reproducibility is the hallmark of the scientific method

- Built for EHT but can be expanded to any project with large data sets
- Combines git logic and pandas organization
- Extracts structured metadata directly from data paths
- Treats datasets as queryable, reproducible objects
- Accessible through Python or CLI interfaces

`hallmark`: Key Features

- Parses parameters from file names to construct a ParaFrame
- Supports Git-like behavior:
  - Branching to maintain and analyze separate data sets
  - Add and index files from structured path patterns
  - Version control through commit history
- Tracks, manages, and recovers exact data sets for reproducible workflows
- All tracked files can be easily read off from, e.g., GitHub:
  - https://github.com/l6a/hallmark-demo-repo.hm/blob/main/data.tsv

hallmark: Key Features

hallmark repo:

```
data/
├── SR1_M87_2017_95_hi_hops_netcal_Stokes.uvfits
├── SR1_M87_2017_95_lo_hops_netcal_Stokes.uvfits
├── SR1_M87_2017_96_hi_hops_netcal_Stokes.uvfits
├── ...
├── .hm
│   ├── .git
│   ├── config.yml
│   ├── data.tsv
│   ├── objects/
```

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hallmark: Key Features

```
fmt: '{release}_{source}_{year}_{doy}_{band}_{pipeline}_{step}_{type}.uvfits'
```

1	sha1	release	source	year	doy	band	pipeline	step	type
2	9eceaa05f7417e8d92009030d7602f0176d39e37	SR1	M87	2017	95	hi	hops	netcal	StokesI
3	da8d1287ddee2ae46a3b0266b3dd11f151de8306	SR1	M87	2017	95	lo	hops	netcal	StokesI
4	87d8be3e4574e556fb1a4741530c426e68b2fab5	SR1	M87	2017	96	hi	hops	netcal	StokesI
5	9d9c95996033145afd1af5011f49c18c8e770259	SR1	M87	2017	96	lo	hops	netcal	StokesI
6	5142cc7510a479607ee48886ba53658cda03b01d	SR1	M87	2017	100	hi	hops	netcal	StokesI
7	c512014cb9ee3ef695507f383939d984756de8a3	SR1	M87	2017	100	lo	hops	netcal	StokesI
8	534882f60ef6af9fc85961019dfccab8cd87774d	SR1	M87	2017	101	hi	hops	netcal	StokesI
9	29ad5b8b3d08e79eff3ea7cbd18e5a31646b944e	SR1	M87	2017	101	lo	hops	netcal	StokesI

`hallmark`: Example Use Cases

- Use `hallmark` CLI to get data from a remote server
  - Example: You create a large simulation dataset on the HPC, put them into a hallmark repo, then upload files to a cloud server. To work with the same dataset in a new local environment, you can clone the hallmark repo, which automatically downloads the data and meta data.
- Use `hallmark` Python API to standardize/normalize inputs for data processing
  - Example: Manipulate ParaFrame
  - Mirrors CLI behavior pythonically

Demos

Closing Statements

- Repo link: https://github.com/l6a/hallmark
- Docs link: https://l6a.github.io/hallmark/
- Pip installable: `$ pip install hallmark`
  - We'll make an updated release of the code you've seen in this demo on Friday
- Please try `hallmark`! We'd love your feedback in the github issues
  - `hallmark` development will continue over the next year. If you have ideas for additional features, please let us know!
- Let CK know if you need interns for software projects

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